

Muhammad Abu Bakar

linkedin.com/in/abu-bakar-ilyas-72b261113/

Karachi, Pakistan

Email : abubakarilyas624@gmail.com

Mobile/WhatsApp : +92-349-1254752

Github : github.com/abubakar12

SUMMARY

AI engineer who ships agentic systems as production software, async-first, high-concurrency, and still running after the handoff. 7+ years across LLM, real-time voice, and generative video, in banking, insurance, healthcare, e-commerce, and hospitality: an agentic-RAG compliance platform built from scratch and live at 8 U.S. banks and credit unions (async FastAPI, SSE streaming, Kubernetes/AKS); a multi-agent harness over MCP tool servers with sub-agent orchestration, skills, and memory; a full-duplex voice assistant with agentic screen control shipped aboard high-end cruise ships; a containerized concept-to-finished-video ad pipeline with an RL-trained verifier in place of the human. Leads senior engineers alongside fleets of coding agents and turns every fix into reusable skills, tools, and eval harnesses.

EXPERIENCE

- The 360factors** USA (Remote)
Oct 2024 – Present
 - *Principal AI Engineer*
 - **Technical Leadership & Quality Bar:** Led a senior, multidisciplinary team of 6–7 (front-end, backend, data, ML); scoped work, reviewed approaches and PRs against a shared system model, and held the enterprise quality bar before anything merged. Partnered directly with the CTO in a tight loop from vision to production.
 - **Enterprise Agentic RAG System (Production, Fortune-class Clients):** Architected and shipped an AI-powered compliance expert system on an agentic RAG architecture with specialized LLMs — defensive coding, validation, edge-case handling, and automated evals throughout (no open-loop AI). Detects compliance issues and surfaces regulatory recommendations against U.S. financial and insurance frameworks; deployed in production for major U.S. banking and insurance customers.
 - **Full-Stack, Agent-Assisted Delivery:** Shipped enterprise features full-stack — from reviewed plan/spec through agent-assisted execution to merged production releases — fast enough to reflect product and customer feedback within the day. Fine-tuned a Llama-3 agent for reliable NL→SQL with strong hallucination guardrails, and integrated Microsoft Fabric for conversational dashboard exploration.
 - **Cloud, CI/CD & MLOps Ownership:** Owned products end to end as a one-person band: containerized (Docker/Kubernetes) training and inference, CI/CD delivery, model versioning, monitoring, and automated retraining (MLflow) — standing services up, shipping, and operating them without handoff.
- Resilia (resilia.shop)** Remote
Jan 2026 – June 2026
 - *Principal AI Engineer/Consultant*
 - **Agent-Owned Pipelines with Self-Verifying Harnesses:** Built an automated ad-generation and research system where agents own whole end-to-end workflows. Fine-tuned an LLM with reinforcement learning as the self-verifying harness that continuously evaluates the pipeline and auto-promotes the best-performing (scaling) ads — pushing the human out of the loop and replacing manual creative testing.
 - **Leverage over Tasks — Reusable Tools, Skills & MCP:** Designed the reusable skills, context, and MCP tools that let a lean team plus its agents deliver at multiples of its size; each class of work needs less intervention every cycle. Operate closed-loop and in parallel across a fleet of frontier coding agents, testing new models/tools continuously rather than locking to one.
 - **End-to-End AI Video Generation Pipeline:** Engineered a fully automated, containerized pipeline generating complete ad videos from only a concept and product description — higher visual consistency and quality than leading tools such as Higgsfield — owning it end to end from infrastructure to shipped output.
- TwinCiti** Remote
Mar 2024 – Feb 2025
 - *Senior AI Engineer*
 - **Real-Time Voice AI — Elsa, a Voice-Native Personal Assistant:** Built Elsa, a personal assistant operated entirely through natural speech and deployed in-cabin aboard high-end cruise ships. Engineered the streaming full-duplex voice loop end to end — ASR, reasoning, and synthesis under conversational latency, with barge-in, interruption recovery, and continuous turn detection — so the assistant is spoken to rather than typed at.
 - **Voice to Action — Agentic Screen Control:** Pushed Elsa past dialogue into execution: spoken intent resolved into grounded control of the on-screen environment, letting guests navigate, launch, and operate applications hands-free. Voice layer and control layer share one agent state, so conversation and action stay in the same loop instead of behind a command grammar.
 - **Agentic Personal Search & a Pluggable Assistant Platform:** Built multi-step agentic search that plans and executes web lookups against each user's own context — preferences, itinerary, personal data — rather than generic queries. Designed it as a pluggable package where a user's services and data attach to the assistant, making Elsa one complete product instead of a bundle of features.

- Afiniti Software Solutions** USA (Remote)
Data Scientist Nov 2022 – Mar 2024
 - **Multi-Agent Orchestration:** Built autonomous multi-agent systems (Microsoft AutoGen) with an orchestrator dynamically instantiating sub-agents for diverse tasks — news retrieval, research-paper analysis, GUI automation (OmniParser + Qwen) — with multi-turn evaluation harnesses for robustness.
 - **Reusable Tooling & Pipeline Modernization:** Led migration of Afiniti’s core ML pipeline from R to Python and overhauled its graph-based call-center pairing pipeline with automated grid search, fold-wise validation, automated deployment, and diagnostic tooling reused across the team. Received the Rising Star Award.
- TheAiSystems** Karachi, Pakistan
Senior Data Scientist Aug 2019 - Nov 2022
 - **Demand Forecasting:** Developed machine learning and statistical models for their demand forecasting pipelines.
 - **Delivering Presentations to Potential and Existing Clients:** Delivering presentations to clients about how the AI can improve their business decisions. This involved showing dashboards, comparing their results vs AI results and showing how interpretability of AI models.
 - **Demand Forecasting Project for Medusa Distributors:** Designed end to end pipeline to predict Medusa’s 8000 different vape based products daily on the 12 week horizon period. This allowed Medusa to reduce their stockout by more than 50%. Algorithm included ensemble forecasts from Tree based and Generalized Linear Models.
 - **Demand Forecasting Project for Friesland Campina (Dutch Dairy Company):** Designed complete demand forecasting pipeline to predict their two products Olpers and Tarang. The accuracy of our forecasts were more than 30% overall over their predictions. This helped saving them millions in revenues.
 - **Demand Forecasting Project for Arcelik:** Led complete forecasting project for the Arcelik. This involved AI forecasts for their 10 different categories comprising more than 400 items. The project also involved providing them with analytics dashboards to help them make improved decisions.

ACHIEVEMENTS

- **Kaggle Competitions:** 1st place / 18,000+ in “Predict Future Sales”; Silver (4,000+) in “GoDaddy Microbusiness Density Forecasting”; Silver (2,500+) in “Learning Agency Lab – Automated Essay Scoring 2.0”.
- **Awards & Projects:** NUST High Achievers Award. Built an autonomous farming robot (NVIDIA autonomous-vehicle research), ~70% edge-detection accuracy. Project Demo

EDUCATION

- Georgia Institute of Technology** Atlanta, USA
Master of Science in Computer Science – GPA: 4.00/4.00 2024
- National University of Sciences and Technology** Islamabad, Pakistan
Bachelor of Engineering in Electrical Engineering – GPA: 3.52/4.00 2016

TECHNICAL SKILLS

- **AI-Native Engineering:** Agentic coding tools & fleet orchestration (Claude Code, Cursor, and frontier alternatives); multi-agent / orchestrator-sub-agent workflows, parallel execution, worktrees; building reusable skills, rules, context, and self-verifying eval harnesses
- **Model Context Protocol (MCP):** Building and consuming MCP tools/servers to give agents context and capability
- **LLMs & Applied AI:** Production LLM integration via APIs, prompt engineering, agentic RAG, fine-tuning (LoRA/QLoRA/PEFT); GPT, Llama-2/3, Qwen; current judgment across frontier model releases and trade-offs
- **Forecasting & Classical ML:** Gradient-boosted ensembles (XGBoost, LightGBM, CatBoost), foundation time-series models (Amazon Chronos), stacking, large-scale feature engineering, anomaly detection
- **Full-Stack & Languages:** Python (expert), SQL, JavaScript/TypeScript, C++, Bash; back-end services, APIs, and data/UI integration
- **Cloud, CI/CD & Delivery:** AWS, Azure; Docker, Kubernetes, CI/CD pipelines, MLflow, Airflow, Git — end-to-end ownership
- **Enterprise Standards:** Defensive coding, validation, security, resilience, automated evals; PR review and architectural decision-making
- **Certifications:** Deep Learning Specialization, NLP Specialization, AWS Cloud Certification (2025)